

CSE3003 OBJECT ORIENTED PROGRAMMING (Fall 2021- 2022)

Project 1: Guessing Games

In this project, you are expected to create a guessing game with different game options. You must design a **player class** and the **GuessGame** base class when creating this game. The player class must have the username, the name of the game played, and the score. GuessGame class should contain the basic functions of the game (**start ()**, **end ()**, **rules ()**, **number of attempts**, **player**, **difficulty level etc.**) Using this **base class**, you need to **design other classes for the 3 games** mentioned below. You must design separate classes for each game. Add **comments for all functions and classes**.

GuessMovie -> In this game you have to determine the names of various movies. When the player wants to play this game, one of the movies you have should be chosen randomly. The number of letters to be hidden must be determined according to the difficulty level that the player will specify at the beginning of the game. E.g. Let's say the randomly chosen movie is "**The Godfather**". If the user has chosen easy as the difficulty level, half the length of the movie should be opened. Like "**T** G*df**he***". If the user wants to play at medium difficulty level, this time a third of the letters should be unlocked. For example, like "**T** **df**h****". If the user wants to play on the hard level, a quarter of the letters must be shown like "**T** G*****h****". After the game starts, the player should have two options:

1. Player can guess the movie name.
2. The player can request a letter.

If the player guesses the name of the Movie correctly, the player's score is updated and the game is over.

If the player wants to get a letter, the first letter hidden in the movie name is unlocked. For example, for the movie that is shown as "**T** G*df**he***" when the game starts, it will be displayed as "**Th* G*df**he***" when a letter is taken. The player may request letters until the movie name is displayed in full. If the player

unlocks all the hidden letters by taking letters without guessing, the game is over. The player does not get any points.

GuessNumber -> If the user chooses the number guessing game, a random number is kept according to the difficulty level set at the beginning of the game. For example, if the player chooses the easy level, the number to be kept must be between 0-20. If he chooses medium level, the number must be between 0-80. If he chooses the hard level it should be between 0-150. Hints should be given according to the number entered by the user. If the guessed number is less than or greater than the selected number, relevant messages should be displayed. For example, "Guess a number greater than your number!" or "Guess a smaller number!". The player's score is calculated based on the number of tries.

GuessCity-> You must have selected various cities of at least 3 countries for this game. When this game is selected, country names is shown to the user. The player chooses from which country player wants to guess the city. After the selection, a city is randomly selected for the specified country. This city is shown with all letters hidden. For example, let's say the selected country is Turkey and the randomly selected city is Istanbul. It should be displayed as "*****". In this game, the player can guess as many times as he/she wants for the easy level, 5 times for the medium level and 2 times for the hard level. The player can take as many letters as he/she wants. Each letter is evaluated according to its difficulty level while calculating the score. For example, each letter taken at easy level can be calculated as -3, for intermediate level as -5, for hard level as -7.

Then the player should have two option:

1. Player can guess letters.
2. The player can guess the city name.

If the player wants to guess the letter, must enter a letter. This letter entered is searched in the city name and the first letter found is displayed. Even if there is more than one letter in the city name, only one letter is shown at a time. For example, if the city name is "**Ankara**", it will be shown as "*****". If the player guessed the letter "**a**", it will be shown again as "**A*******". The game continues like this. Score is calculated according to the attempts and the game is over.

A **menu** should be shown when the player **starts the game**. The **games** mentioned above should **be listed in this menu**. After the user selects the game, **the game rules** should be explained. After the game rules, the player should be asked to determine the difficulty level. The selected game starts. When the **game is over**, the **user name and score** are **displayed** on the screen.

Important Note: Do not share your solution with others! Do not collaborate with each other! You have to do your project by yourself. Otherwise, your submission will not be evaluated and both you and the person that you get the file from will get zero points from the project.